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Zekun Wang

zekun@gatech.edu · (814) 699-1471 · ndrsn0208.github.io · Google Scholar

Research Focus

I study continual learning for deployment-time adaptation and generalization in foundation models and AI agents. I investigate how these systems can learn from ongoing experience, adapt to unfamiliar tasks and environments, and improve over time while preserving prior knowledge and capabilities.

Education

Georgia Institute of Technology *2024 – 2029*
Ph.D. in Computer Science
Advisor: Christopher J. MacLellan

University of Michigan *2022 – 2024*
M.S. in Computer Science

Pennsylvania State University *2018 – 2022*
B.S. in Computer Science
Double major: Mathematics. Minor: Economics.

Industry Experience

Amazon *08/2026 – present*
Research Intern · WA, USA

- **One-shot adaptation to novel tasks.** Developing methods that adapt an agent to an unfamiliar task at deployment through a single update to its harness (LeAH, preprint).
- **Self-improving agent harnesses.** Contributing to a harness self-improvement system that outperforms baseline harness-evolution methods on SWE-bench and Terminal-Bench.

Selected Publications

For the full publication list, see Google Scholar. * Equal contribution.

Trust Region Continual Learning as an Implicit Meta-Learner *NeurIPS 2026*
Zekun Wang*, Anant Gupta*, Christopher J. MacLellan

- Proved that replay with a Fisher-metric trust region yields a one-step MAML-like update under local approximations, connecting continual learning to meta-learning.
- Improved retention and re-convergence over EWC, replay, FTML, and VR-MCL on ImageNet-500 diffusion generation and Continual-World-10 diffusion policies.

LeAH: Generalizing Test-Time Agent Harness Adaptation to Unseen Environments *Preprint*
Zekun Wang, Yingqiang Ge, Hai Wang, Yuhui Xu, Joshua Frandsen, Eric Yang, Shancong Fu, Christopher J. MacLellan, Chandan K. Reddy

- Formulated harness adaptation that generalizes to new benchmarks as a joint objective and meta-trained a 9B proposer that reads a few failures of an unseen benchmark and writes one harness edit, **without search or weight updates**.
- Raised the average score on six unseen benchmarks from 51.4 under the initial harness to 57.9 with a frozen agent. Work done during an internship at Amazon.

Self-Consolidating Language Models *Preprint*
Zekun Wang[*], Anant Gupta[*], Zihan Dong, Christopher J. MacLellan

- Developed a meta-RL post-training framework in which an LLM learns which layers to update as new context arrives, balancing knowledge acquisition against forgetting.
- Improved both acquisition and retention on SQuAD over every evaluated baseline; meta-trained only on short LongBench v2 contexts, it generalized to longer contexts and outperformed in-context methods there.

Test-Time Compositional Generalization in Diffusion Models via Prototype Discovery *Preprint*
Zekun Wang, Anant Gupta, Tianyi Zhu, Christopher J. MacLellan

- Developed test-time prototype discovery in pretrained diffusion models: score ascent at several noise levels recovers prototypes **without labels, a concept library, or additional training**.
- Composed the prototypes that explain a single out-of-distribution query into a product of experts; on held-out ColorMNIST and CelebA compositions, samples generalized to the unseen class, where sampling from trained concept tokens landed on neighboring classes.

Avoid Catastrophic Forgetting with Rank-1 Fisher from Diffusion Models *ICLR 2026*
Zekun Wang[*], Anant Gupta[*], Zihan Dong, Christopher J. MacLellan

- Derived and empirically validated a low-SNR rank-1 Fisher approximation near convergence, enabling structured EWC with generative replay.
- Reduced mean FID-based forgetting by **67% on ImageNet-1k** (32×32, 20 tasks) versus generative distillation alone; nearly eliminated forgetting on MNIST and FashionMNIST.

Deep Taxonomic Networks for Unsupervised Hierarchical Prototype Discovery *NeurIPS 2025*
Zekun Wang, Ethan Haarer, Tianyi Zhu, Zhiyi Dai, Christopher J. MacLellan

- Developed a hierarchical generative model that discovers prototypes without a known class count and achieved state-of-the-art hierarchical prototype discovery in the paper's evaluation.
- Proved that its variational objective also optimizes mutual information between latent representations and the hierarchy.

Honors and Awards

Ada Lovelace Award for Excellent Paper *2026*
Advances in Cognitive Systems (ACS)
For Diffusion Models and Concept Formation.

Patent Applications

Systems and Methods for Continual Learning of Diffusion Models
U.S. Patent Application No. 19/821,778, filed September 28, 2026.
Co-inventor. Applicant: Georgia Tech Research Corporation.

System and Method for Deep Taxonomic Networks for Unsupervised Hierarchical Prototype Discovery and Classification
U.S. Patent Application No. 19/822,642, filed September 28, 2026.
Co-inventor. Applicant: Georgia Tech Research Corporation.

Method and Apparatus for Taxonomic Networks with Neuro-Symbolic Pairing
U.S. Patent Application No. 19/694,309, filed June 1, 2026.
Co-inventor. Applicant: Georgia Tech Research Corporation.

Information-Theoretic Hierarchical Concept Learning for Masked-Word Prediction
U.S. Patent Application No. 19/237,713, filed June 13, 2025.
Co-inventor. Applicant: Georgia Tech Research Corporation.

Research Experiences

Georgia Institute of Technology *Aug. 2024 – present*
Graduate Research Assistant — Teachable AI Lab

University of Michigan *2022 – 2024*
Graduate Research Assistant — SLED Lab (PI: Joyce Chai)

Pennsylvania State University *2021 – 2022*
Undergraduate Research Assistant

Teaching Experiences

Graduate Student Instructor — EECS 595, Natural Language Processing *Sept. 2023 – May 2024*
University of Michigan

Teaching Assistant — CMPSC 464, Introduction to the Theory of Computation *Jan. 2022 – May 2022*
Pennsylvania State University

Teaching Assistant — MATH 230, Calculus and Vector Analysis *Aug. 2021 – Dec. 2021*
Pennsylvania State University

Math Peer Tutor — Calculus *Jan. 2020 – Jan. 2022*
Penn State Learning

Invited Talks and Guest Lectures

Intro to AI (CS 3600), Georgia Tech — Guest lecture *Aug. 2025*

Deep Learning: Classics and Trends (DLCT) — Invited talk *July 2024*
Babysit a Language Model From Scratch: Interactive Language Learning by Trials and Demonstrations

Advanced Artificial Intelligence (EECS 692), U. of Michigan — Guest lecture *Jan. 2023*

Academic Leadership and Service

Workshop Organizer *Dec. 2026 (upcoming)*
Towards Test-Time Continual Learning Agents (TTCL), NeurIPS 2026

Student Volunteer: ACS.

Conference Reviewing: NeurIPS, ICML, ICLR, AAAI, ACL, NAACL, ACS.

Research Mentoring

Anant Gupta: M.S., Georgia Tech. 2025 – present.
Karthik Singaravadivelan: B.S., Georgia Tech. 2025 – present.
Aryan Roy: M.S., Georgia Tech. 2025 – present.
Zihan Dong: M.S., Georgia Tech. / Lenovo. 2024 – present.
Ethan Haarer: M.S., Georgia Tech. 2024 – 2025.

Skills

Programming	Python, C/C++, CUDA, R, MATLAB, Java, C#, TEX, SQL
ML Frameworks	PyTorch, Hugging Face (Transformers, Diffusers, PEFT/LoRA, TRL), Accelerate, DeepSpeed, vLLM, JAX (working), Weights & Biases
Infrastructure	SLURM, Docker, multi-node A100/H100 clusters, Linux/Bash, Git
Mathematics	Information / probability theory, Bayesian inference, calculus, linear algebra, variational methods, score matching, optimization
Research areas	Continual / lifelong learning, LLM post-training (RLHF/PPO/DPO/meta-RL), diffusion & flow models, meta-learning, compositional generalization, concept formation
Languages	English (fluent), Cantonese (fluent), Mandarin Chinese (native)

Full Publication List

Trust Region Continual Learning as an Implicit Meta-Learner *2026*
Zekun Wang[*], Anant Gupta[*], Christopher J. MacLellan
NeurIPS 2026

LeAH: Generalizing Test-Time Agent Harness Adaptation to Unseen Environments *2026*
Zekun Wang, Yingqiang Ge, Hai Wang, Yuhui Xu, Joshua Frandsen, Eric Yang, Shancong Fu, Christopher J. MacLellan, Chandan K. Reddy
Preprint

Diffusion Models and Concept Formation *2026*
Zekun Wang, Karthik Singaravadivelan, Christopher J. MacLellan
Advances in Cognitive Systems 2026
Ada Lovelace Award for Excellent Paper

VERSE: Verified Self-Evolving Optimizer for Agent Harnesses *2026*
Zekai Wang, Yingqiang Ge, **Zekun Wang**, Hai Wang, Yuhui Xu, Joshua Frandsen, Shancong Fu, Ashia C. Wilson, Chandan K. Reddy
arXiv preprint

A Rational Account of Categorization Based on Information Theory *2026*
Christopher J. MacLellan, Karthik Singaravadivelan, Xin Lian, **Zekun Wang**, Pat Langley
CogSci 2026

Decompose Sparsely Where You Should, Absorb Densely Where You Should Not *2026*
Ruixuan Deng, Zehao Jin, **Zekun Wang**, Zihan Dong
ICML 2026 Mechanistic Interpretability Workshop (Poster)

CobwebTM: Probabilistic Concept Formation for Lifelong and Hierarchical Topic Modeling *2026*
Karthik Singaravadivelan[*], Anant Gupta[*], **Zekun Wang**, Christopher J. MacLellan
Findings of ACL 2026

Grounded Concreteness: Human-Like Concreteness Sensitivity in Vision-Language Models *2026*
Aryan Roy, **Zekun Wang**, Christopher J. MacLellan
Findings of ACL 2026

Self-Consolidating Language Models *2026*
Zekun Wang[*], Anant Gupta[*], Zihan Dong, Christopher J. MacLellan
Preprint

Test-Time Compositional Generalization in Diffusion Models via Prototype Discovery *2026*
Zekun Wang, Anant Gupta, Tianyi Zhu, Christopher J. MacLellan
Preprint

Avoid Catastrophic Forgetting with Rank-1 Fisher from Diffusion Models *2026*
Zekun Wang[*], Anant Gupta[*], Zihan Dong, Christopher J. MacLellan
ICLR 2026

Computer Vision Modeling of the Development of Geometric and Numerical Concepts in Humans *2026*
Zekun Wang, Sashank Varma
AAAI 2026

Deep Taxonomic Networks for Unsupervised Hierarchical Prototype Discovery *2025*
Zekun Wang, Ethan Haarer, Tianyi Zhu, Zhiyi Dai, Christopher J. MacLellan
NeurIPS 2025

Hierarchical Semantic Retrieval with Cobweb *2025*
Anant Gupta[*], Karthik Singaravadivelan[*], **Zekun Wang**
Advances in Cognitive Systems 12 (2025)

Efficient and Scalable Masked Word Prediction Using Concept Formation *2025*
Xin Lian[*], **Zekun Wang**[*], Christopher J. MacLellan
Cognitive Systems Research 92, 101371

Computer Vision Models Show Human-Like Sensitivity to Geometric and Topological Concepts *2025*
Zekun Wang, Sashank Varma
CogSci 2025

Taxonomic Networks: A Representation for Neuro-Symbolic Pairing *2025*
Zekun Wang, Ethan L. Haarer, Nicki Barari, Christopher J. MacLellan
NeuS 2025

Babysit A Language Model From Scratch: Interactive Language Learning by Trials and Demonstrations *2025*
Ziqiao Ma[*], **Zekun Wang**[*], Joyce Chai
NAACL 2025

Has It All Been Solved? Open NLP Research Questions Not Solved by Large Language Models *2024*
Oana Ignat[*], Zhijing Jin[*], Artem Abzaliev, Laura Biester, Santiago Castro, Naihao Deng, Xinyi Gao, Aylin Gunal, Jacky He, Ashkan Kazemi, Muhammad Khalifa, Namho Koh, Andrew Lee, Siyang Liu, Do June Min, Shinka Mori, Joan Nwatu, Veronica Perez-Rosas, Siqi Shen, **Zekun Wang**, Winston Wu, Rada Mihalcea
LREC-COLING 2024

Foundations for AI-Assisted Formative Assessment Feedback for Short-Answer Tasks in Large-Enrollment Classes *2022*
Susan E. Lloyd, Matthew D. Beckman, Dennis K. Pearl, Rebecca J. Passonneau, Zhaohui Li, **Zekun Wang**
ICOTS 2022